Ramp It Up 2.0

Chapter 3 · REQUIRED · Tuesday and Wednesday

Name _____ Date _____

QUESTION: How does changing ramp height affect the change caused by the same ball?

PREDICT: Which ramp height do you predict will move the cup farthest? Why?

_____

FAIR TEST — keep the same:

· same ball · same ramp · same release spot · let go, do not push

· same cup · same cup starting spot · same measuring method

CHANGE ONE THING: ramp height. MEASURE: ramp height (cm) and cup movement (cm).

Roll once to calibrate. Do not record that roll. Then record three trials at each height. That is nine measured trials.

Ramp height is not a speed. Do not write a speed number.

Condition / books	Actual ramp height (cm)	Trial 1 cup movement (cm)	Trial 2 cup movement (cm)	Trial 3 cup movement (cm)	Average cup movement (cm)
1 book					
2 books					
3 books					

GRAPH: Plot your three averages. Do not label an axis as speed.

y-axis — Average cup movement (cm)

x-axis — Actual ramp height (cm)

Challenge: move the cup as close as you can to 30 cm without going past.

My closest result: _____ cm. Did it go past 30 cm? yes / no

READ THE DATA: Which ramp height had the greatest average cup movement?

_____

COMPARE: Compare your 1-book and 3-book averages. Which moved the cup farther?

_____

NOTICE THE PATTERN: As ramp height increased, what happened to the average cup movement?

_____

CONNECT TO SPEED: The Reader says objects pick up more speed on steeper ramps. Which setup gave the same ball more speed?

_____

CONNECT TO ENERGY: The Reader says the faster the ball is moving, the greater its energy of motion. Which setup gave the same ball more energy of motion?

_____

CLAIM: When the same object moves faster, it has _____ energy of motion.

EVIDENCE: Write TWO actual average cup-movement values from your table that support your claim.

_____ and _____

REASONING: Explain how those two values support your claim.

_____

_____

EXTEND YOUR THINKING: If you tested a ramp height between two of your tested heights, what cup movement would you predict? Explain using your graph.

_____